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Degree: B.Tech

Roll no.:
Semester: 3

Mid Semester Examination, September 2023

Course Code: EIMTC04/ITMTC04/INMTC04

Course title: Optimization principles and techniques

Time: 1 hour 30 minutes

Maximum Marks: 25

Note: Missing data/information(if any), may be suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO																														
1a	A manufacturing company makes two models A and B of a product. Each piece of Model A requires 9 labour hours for fabricating and 1 labour hour for finishing. Each piece of Model B requires 12 labour hours for fabricating and 3 labour hours for finishing. For fabricating and finishing, the maximum labour hours available are 180 and 30 respectively. The company makes a profit of Rs 8000 on each piece of model A and Rs 12000 on each piece of Model B. How many pieces of Model A and Model B should be manufactured per week to realise a maximum profit? Formulate this problem as LPP.	2.5	CO1																														
1b	Solve the above formulated LPP graphically and discuss the solution.	2.5	CO1																														
2a	Solve by Simplex method: $\begin{aligned} \text{Max } Z &= 3x_1 + 2x_2 + 5x_3 \\ \text{subject to} \\ x_1 + 2x_2 + x_3 &\leq 430 \\ 3x_1 + 2x_3 &\leq 460 \\ x_1 + 4x_2 &\leq 420 \\ x_i &\geq 0 \end{aligned} \quad (1)$	2.5	CO1																														
2b	Without sketching the graph, find all the vertices of the set of the feasible solutions of : $\begin{aligned} -x_1 + x_2 &\leq 1 \\ 2x_1 - x_2 &\leq 2 \\ x_1 \geq 0, x_2 &\geq 0 \end{aligned}$	2.5	CO1																														
3a	Write the dual of the following LPP: $\begin{aligned} \text{Max } Z &= 5x_1 - 4x_2 + 6x_3 \\ \text{subject to} \\ 2x_1 + 6x_2 + 4x_3 &\leq 7 \\ 6x_1 - x_2 + 3x_3 &\leq 4 \\ 5x_1 + 2x_2 + x_3 &\geq 10 \\ x_1 - 2x_2 + 5x_3 &\geq 3 \\ 2x_1 + 5x_2 - 2x_3 &= 5 \\ x_1 \text{ unrestricted, } x_2 \leq 0, x_3 &\geq 0 \end{aligned}$	2.5	CO2																														
3b	The following table gives the unit transportation costs, the supply at each origin and the demand of each destination for a transportation problem <table><tr><td></td><td>D1</td><td>D2</td><td>D3</td><td>D4</td><td>Availability</td></tr><tr><td>O1</td><td>3</td><td>4</td><td>8</td><td>7</td><td>60</td></tr><tr><td>O2</td><td>7</td><td>3</td><td>7</td><td>6</td><td>80</td></tr><tr><td>O3</td><td>3</td><td>9</td><td>3</td><td>4</td><td>100</td></tr><tr><td>Demand</td><td>40</td><td>70</td><td>50</td><td>80</td><td></td></tr></table> Let x_{ij} denote the number of units to be transported from origin i to destination j. If the uv method is applied to improve the basic feasible solution given by $x_{12} = 60, x_{22} = 10, x_{23} = 50, x_{24} = 20, x_{31} = 40, x_{34} = 60$, then find the optimal cost of transportation.		D1	D2	D3	D4	Availability	O1	3	4	8	7	60	O2	7	3	7	6	80	O3	3	9	3	4	100	Demand	40	70	50	80		2.5	CO2
	D1	D2	D3	D4	Availability																												
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Demand	40	70	50	80																													

4a	State weak and strong duality theorems. Discuss the relationship between the solutions of primal and dual LPPs.	2.5	CO2																														
4b	If $\{x_{13} = a, x_{22} = b, x_{31} = c, x_{32} = d, x_{34} = e, x_{23} = 10\}$ is the set of basic variables of a balanced transportation problem seeking to minimize cost of transportation from origins to destinations, where the cost matrix is <table><tr><td></td><td>D1</td><td>D2</td><td>D3</td><td>D4</td><td>Availability</td></tr><tr><td>O1</td><td>6</td><td>2</td><td>-1</td><td>0</td><td>10</td></tr><tr><td>O2</td><td>4</td><td>2</td><td>2</td><td>3</td><td>$\lambda+5$</td></tr><tr><td>O3</td><td>3</td><td>1</td><td>2</td><td>1</td><td>3λ</td></tr><tr><td>Demand</td><td>10</td><td>$\mu-5$</td><td>$\mu+5$</td><td>15</td><td></td></tr></table> where $\lambda, \mu \in \mathbb{R}$. Find the values of a,b,c, d,e.		D1	D2	D3	D4	Availability	O1	6	2	-1	0	10	O2	4	2	2	3	$\lambda+5$	O3	3	1	2	1	3λ	Demand	10	$\mu-5$	$\mu+5$	15		2.5	CO2
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O3	3	1	2	1	3λ																												
Demand	10	$\mu-5$	$\mu+5$	15																													
5a	We have to assign four jobs, I, II, III, IV to four workers A, B, C and D. The time taken by different workers (in hours) in completing different jobs is given below: <table><tr><td></td><td>I</td><td>II</td><td>III</td><td>IV</td></tr><tr><td>A</td><td>5</td><td>3</td><td>2</td><td>8</td></tr><tr><td>B</td><td>7</td><td>9</td><td>2</td><td>6</td></tr><tr><td>C</td><td>6</td><td>4</td><td>5</td><td>7</td></tr><tr><td>D</td><td>5</td><td>7</td><td>7</td><td>8</td></tr></table> Find the minimum time (in hours) taken by the workers to complete all jobs.		I	II	III	IV	A	5	3	2	8	B	7	9	2	6	C	6	4	5	7	D	5	7	7	8	2.5	CO2					
	I	II	III	IV																													
A	5	3	2	8																													
B	7	9	2	6																													
C	6	4	5	7																													
D	5	7	7	8																													
5b	Use dual simplex method to solve the following LPP: Min $Z = 5x_1 + 6x_2$ subject to $x_1 + x_2 \geq 2$ $4x_1 + x_2 \geq 4$ $x_1 \geq 0, x_2 \geq 0$	2.5	CO2																														